PHY203 - Mechanics Lab

Bachelor, Semester 3
Ecole Polytechnique

Rotating frames

In this lab, we study the effect of working in a non-Galilean reference frame, specifically a rotating one. All experiments are performed on a rotating platform, with an embedded camera that grabs the position and motion of objects in this reference frame. We deduce the equivalent inertial forces created by the continuous rotation at angular speed Ω on still and moving objects, and recreate a meteorological phenomenon in the lab.

1 Dimensional analysis

In all this lab, we only consider the motion in the horizontal plane, in a frame rotating around the vertical axis (unit vector $\mathbf{e}_z$) at a constant angular speed $\mathbf{\Omega} = \Omega \mathbf{e}_z$. We admit that Newton's second law is unchanged along the vertical axis and look only for inertial forces $\mathbf{F}_i$ in the horizontal plane. We consider a punctual object of mass m, position $\mathbf{x}$ and speed $\mathbf{v}$.

1.1. Identify the 5 quantities describing the system. For each, precise their units in the MKS system. From there, show that you can construct only 2 dimensionless quantities both involving the norm of the inertial force and the angular speed. **Here, you will exclude the orientation of the vectors.**

Solution: The five quantities which describe the system are:

1. mass m unit: kg
2. position, or radial distance $r = ||\vec{x}||$ unit: m
3. speed $v = ||\vec{v}||$ unit: $m \cdot s^{-1}$
4. angular speed $\omega = ||\vec{\omega}||$ unit: s^{-1}
5. norm of the inertial force $F_i = ||\vec{F_i}||$ unit: $kg \cdot m \cdot s^{-2}$

Since there are only 3 MKS units involved (kg,m,s), from Buckingham π argument we only need $5 - 3 = 2$ dimensionless quantities to represent the system. If both have to involve the norm of the inertial force and the angular speed, we naturally find:

$$\Pi_1 = \alpha 1 \cdot \frac{F_i}{\omega}, \Pi_2 = \alpha 2 \cdot \frac{F_i}{\omega}$$

to oppose the units of time in F_i with those in ω. Similarly with m and F_i:

$$\Pi_1 = \alpha 3 \cdot \frac{F_i}{m\omega}, \Pi_2 = \alpha 4 \cdot \frac{F_i}{m\omega}$$

As m is the only quantity with units (kg). Finally, we want to use both r and v once, and a quick dimensional analysis yields:

$$\Pi_1 = \frac{F_i}{m\omega^2 r}, \Pi_2 = \frac{F_i}{m\omega v}$$

We now include the orientation of the vectors. We admit that the inertial forces $\mathbf{F}_i$ can be decomposed into 2 quantities: $\mathbf{F}_i = \mathbf{F}_x + \mathbf{F}_v$, with $\mathbf{F}_x$ the centrifugal force depending only on position $\mathbf{x}$, and $\mathbf{F}_v$ the Coriolis force depending only on speed $\mathbf{v}$.

1.2. From the scaling of previous question, deduce what happens to $\mathbf{F}_x$ and $\mathbf{F}_v$ if we change $\mathbf{\Omega}$ to $-\mathbf{\Omega}$.

Solution: From Π_1, Π_2, we find : $\Pi_1 \sim F_x, \Pi_2 \sim F_v \implies F_x = \alpha_x m\omega^2 x, F_v = \alpha_v m\omega v$
Hence: $\omega \to -\omega \implies F_{x,-\omega} = F_{x,\omega}, F_{v,-\omega} = -F_{v,-\omega}$

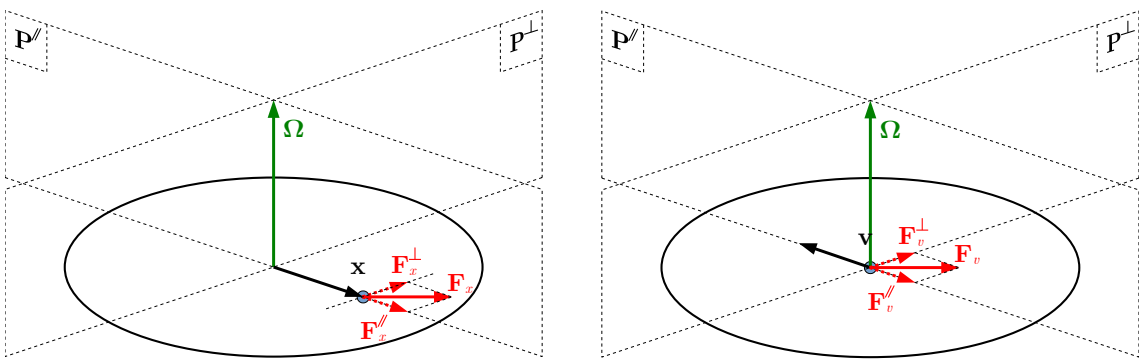


Figure 1: We decompose $\mathbf{F}_x$ in $\mathbf{F}_x^{/\!/} + \mathbf{F}_x^{\perp}$, where $\mathbf{F}_x^{/\!/}$ is parallel to $\mathbf{x}$ and $\mathbf{F}_x^{\perp}$ is perpendicular to $\mathbf{x}$, and $\mathbf{F}_v = \mathbf{F}_v^{/\!/} + \mathbf{F}_v^{\perp}$, where $\mathbf{F}_v^{/\!/}$ is parallel to $\mathbf{v}$ and $\mathbf{F}_v^{\perp}$ is perpendicular to $\mathbf{v}$.

1.3. We decompose $\mathbf{F}_x = \mathbf{F}_x^{/\!/} + \mathbf{F}_x^{\perp}$, where $\mathbf{F}_x^{/\!/}$ is parallel to $\mathbf{x}$ and $\mathbf{F}_x^{\perp}$ is perpendicular to $\mathbf{x}$, see Fig. 1. How are $\mathbf{\Omega}$, $\mathbf{x}$, $\mathbf{F}_x^{/\!/}$ and $\mathbf{F}_x^{\perp}$ modified by reflection perpendicular to the plane $\mathbf{P}^{/\!/}$ defined by vectors $\mathbf{e}_z$ and $\mathbf{x}$? For Ω, make sure to consider the direction of the rotation upon reflection. Conclude that $\mathbf{F}_x$ and $\mathbf{x}$ are aligned, and give the vectorial expression of $\mathbf{F}_x$ as a function of $\mathbf{x}$ up to an unknown numerical factor α_x that we will extract from the experiments.

Solution: We are interested in the reflection by the plane $P^{\perp}$ of the vectors $\vec{\omega}, \vec{x}, \vec{F_x^{/\!/}}$, and $\vec{F_x^{\perp}}$. In backwards order:
Since $\vec{F_x^{\perp}}$ belongs to $P^{\perp}$, it is not changed by reflection, hence $F_2\vec{x}^{\perp} = \vec{F_x^{\perp}}$;
Since $\vec{F_x^{/\!/}}$ is perpendicular to $P^{\perp}$, we have $F_2\vec{x}^{/\!/} = -\vec{F_x^{/\!/}}$
Since $\vec{x}$ is perpendicular to $P^{\perp}$, we have $\vec{x_2} = -\vec{x}$;
Since ω belongs to $P^{\perp}$, it is not changed by reflection, hence $\omega_2 = \omega$. However, by reflection across a plane perpendicular to the disk and which intersects its center, we have that counter-clockwise rotation becomes clockwise rotation and vice-versa. Hence $\vec{\omega_2} = -\vec{\omega}$.
From $F_x = \alpha_x m\omega^2 x$, we find : $R^{\perp}(F_x) = -F_x$, and hence $\vec{F_x} = \vec{F_x^{/\!/}}$, and $\vec{F_x} = \alpha_x m\omega^2 \vec{x}$

1.4. We finally decompose $\mathbf{F}_v = \mathbf{F}_v^{/\!/} + \mathbf{F}_v^{\perp}$, where $\mathbf{F}_v^{/\!/}$ is parallel to $\mathbf{v}$ and $\mathbf{F}_v^{\perp}$ is perpendicular to $\mathbf{v}$. How are $\mathbf{\Omega}$, $\mathbf{v}$, $\mathbf{F}_v^{/\!/}$ and $\mathbf{F}_v^{\perp}$ modified by reflection perpendicular to the plane $\mathbf{P}^{/\!/}$ defined by vectors $\mathbf{e}_z$ and $\mathbf{v}$? Conclude that $\mathbf{F}_v$ and $\mathbf{v}$ are perpendicular, and give the vectorial expression of $\mathbf{F}_v$ as a fonction of $\mathbf{v}$ up to an unknown numerical factor α_v that we will extract from the experiments.

Solution: We are interested in the reflection by the plane $P^{\perp}$ of the vectors $\vec{\omega}, \vec{v}, \vec{F_v^{/\!/}}$, and $\vec{F_v^{\perp}}$. In backwards order:
Since $\vec{F_v^{\perp}}$ belongs to $P^{\perp}$, it is not changed by reflection, hence $\vec{F_2 v^{\perp}} = \vec{F_v^{\perp}}$;
Since $\vec{F_v^{/\!/}}$ is perpendicular to $P^{\perp}$, we have $\vec{F_2 v^{/\!/}} = -\vec{F_v^{/\!/}}$
Since $\vec{v}$ is perpendicular to $P^{\perp}$, we have $\vec{v_2} = -\vec{v}$;
Since ω belongs to $P^{\perp}$, it is not changed by reflection, hence $\omega_2 = \omega$. However, by reflection across a plane perpendicular to the disk and which intersects its center, we have that counter-clockwise rotation becomes clockwise rotation and vice-versa. Hence $\vec{\omega_2} = -\vec{\omega}$.
From $F_v = \alpha_v m \omega v$, we find : $R^{\perp}(F_v) = -(-F_v) = F_v$, and hence $\vec{F_v} = \vec{F_v^{\perp}}$, and $\vec{F_v} = \alpha_v m \vec{\omega} \times \vec{v}$

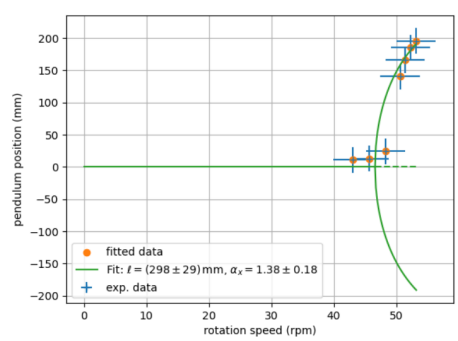
3 Centrifugal force: pendulum experiment

Static equilibrium positions

3.5. Give the value and uncertainty of the critical rotation speed Ω_c where the vertical position gets unstable.

Solution: We found $\Omega_C = 34 \pm 1$ rotations per minute to be when the vertical position becomes unstable, though it is hard to tell with certainty when the system is stable or not.

3.6. Plot the equilibrium positions of the pendulum as a function of the rotation speed. Can you see different equilibrium positions for a given rotation speed? Explain why you can or cannot.



Solution:
Graph of the distance of the bob from the axis of rotation as a discrete function of rotation speed.

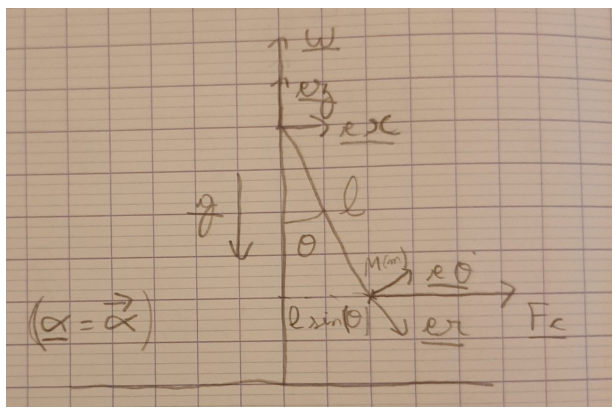
We cannot see different equilibrium positions for a given rotation speed, because either the rotation speed is too low for the pendulum to reach an angled equilibrium, and it is in equilibrium

at the center, or the rotation speed is high enough for the centrifugal force to angle the equilibrium position sideways in the rotating frame of reference, or, as the rotation speed is in between the two, the system will be unstable, and will start angling away from the center as centrifugal force takes effect, but will be brought back down by gravity, in a loop.
Essentially, the system is either unstable or has a unique stable equilibrium.

3.7. Theoretical analysis: using the force or momentum balance between gravity and centrifugal force, show that the horizontal position x of the pendulum's end mass should follow the following equation:

$$x^2 = \ell^2 - \left(\frac{g}{\alpha_x \Omega^2}\right)^2,$$

where ℓ is the length of the pendulum, and g the acceleration of gravity. What approximations have been done to reach this equation? What is the expected expression of Ω_c? Can you extract a value for α_x from your previous measurement of Ω_c? Show that using a smart linear fit of the data, you can estimate α_x and ℓ independently of Ω_c.



Solution:
Since the pendulum is not moving translationally relative to the inertial frame, and at equilibrium, $v = 0 \implies F_{coriolis} = 0, \dot{\Omega} = 0 \implies F_{Euler} = 0$, so the only forces we are left with in the rotating frame of reference are $\vec{F_g} = -mg\vec{e_z}$ and $\vec{F_c} = m\vec{\omega} \times \vec{\omega} \times \vec{r}$, resp. the gravitational force and the centrifugal force.
We quickly find that:

$$\vec{F_g} = -mg\vec{e_z}$$

$$\vec{r} = l\vec{e_r} \implies \vec{F_c} = m\Omega^2 l \sin\theta \vec{e_x}$$

with:

$$\vec{e_x} = \cos\theta\vec{e_\theta} + \sin\theta\vec{e_r}$$

$$\vec{e_z} = \sin\theta\vec{e_\theta} - \cos\theta\vec{e_r}$$

We are interested in equilibrium, which is guaranteed on $\vec{e_r}$ by the rigid pendulum, but along $\vec{e_\theta}$ gives: $\vec{F_g} \cdot \vec{e_\theta} + \vec{F_c} \cdot \vec{e_\theta} = 0$. We find:

$$(m\sin\theta)\Omega^2 l\cos\theta - (m\sin\theta)g = 0$$

$\cos\theta = \dfrac{g}{\Omega^2 l}$, and if we ignore $\theta = 0 \iff \sin\theta = 0$, we have:

$$\cos^2\theta = \left(\frac{g}{\Omega^2}\right)^2 \cdot \frac{1}{l^2}$$

Finally, from $x = l\sin\theta$, we find:

$$x^2 = l^2 \sin^2\theta$$

$$x^2 = l^2 \cdot (1 - \cos^2\theta)$$

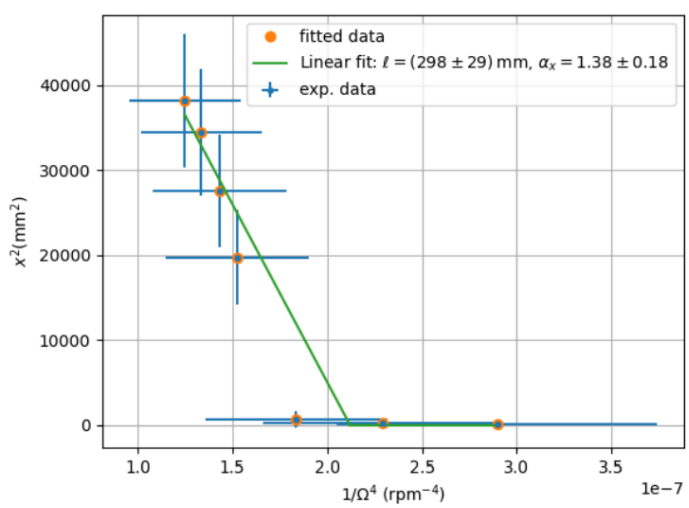
$$x^2 = l^2 - l^2 \cdot \left(\frac{g}{\Omega^2}\right)^2 \cdot \frac{1}{l^2}$$

$$x^2 = l^2 - \left(\frac{g}{\alpha_x \Omega^2}\right)^2$$

To reach this equation, we made the assumption the rod was massless and perfectly rigid, with a time steady equilibrium, no drag or external forces, and perfectly radial tension.
We have $0 \leq \cos\theta = \frac{g}{\Omega^2 l} \leq 1$. No matter the value of Ω, $\theta = \sin\theta = 0$ is a trivial solution. If $\Omega \sim 0$, then $\frac{g}{\Omega^2 l} >> 1$, the only solution is $\theta = 0$. As Ω increases, it will reach a threshold value for which $\frac{g}{\Omega^2 l}$ enters the $[0, 1]$ interval. Hence $\frac{g}{\Omega^2_{\text{thresh}} l} = 1$. The expected expression of $\Omega_C = \Omega_{\text{thresh}}$ is $\Omega_C = \sqrt{\frac{g}{l}}$, or, keeping α_x in the equation, $\Omega_C = \sqrt{\frac{g}{\alpha_x l}}$, and we clearly have $\alpha_x = 1$

3.8. Can you give an estimation (and uncertainty) for α_x, ℓ? Superpose the theoretical prediction to the experimental data and comment on the agreement or disagreement. How does the value of α_x compares with the theoretical one $\alpha_x = 1$?

Solution: From $x^2 + (\frac{g}{\alpha_x \Omega^2})^2 = l^2$,we find that for large Omega, $x^2 \to l^2$. We can use: $x^2 = l^2 - \frac{g^2}{\alpha_x^2} \cdot \frac{1}{\Omega^4}$ which is a straight line in x^2 over $1/\Omega^4$. The intercept is l^2 and the slope is $-g^2/\alpha_x^2$.
Graph of x^2 vs $1/\Omega^4$.

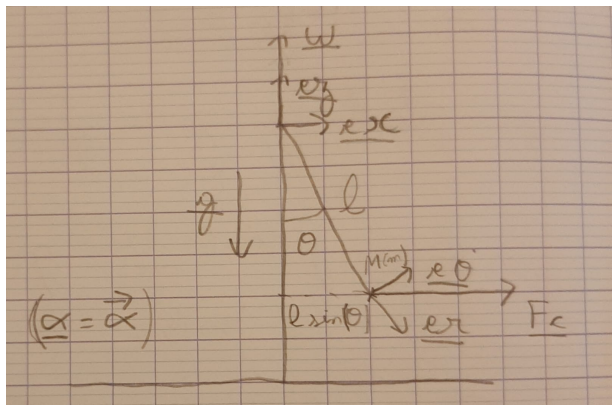


From the linear fit on this graph, we have $l = 298 \pm 29$mm, $\alpha_x = 1.38 \pm 0.18$.

This is rather distant from the theoretical $\alpha_x = 1$, with a standard error of 38%.

Dynamic study of the vertical pendulum

3.9. Theoretical analysis: using Newton's second law applied to the pendulum, for small amplitude oscillations (hence, small angle approximation), show that the motion is that of an harmonic oscillator of angular resonance frequency ω such that:
$\omega^2 = \alpha_x(\Omega_c^2 - \Omega^2)$.

Solution: 

$$\sum \vec{F_\theta} = m\Omega^2 l \sin(\theta)\cos(\theta) - mg\sin(\theta)$$

$$a_\theta = l\ddot{\theta} = \Omega^2 l \sin(\theta)\cos(\theta) - g\sin(\theta)$$

For $\theta_0 \approx 0, \sin(\theta) \approx \theta, \cos(\theta) \approx 1$:

$$l\ddot{\theta} = \Omega^2 l\theta - g\theta$$

$$\ddot{\theta} + (\frac{g}{l} - \Omega^2)\theta = 0$$

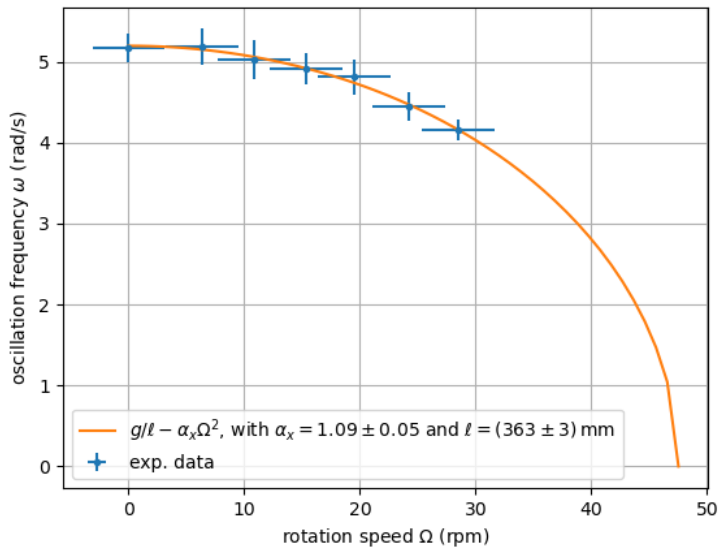
$$\omega^2 = \frac{g}{l} - \omega^2$$

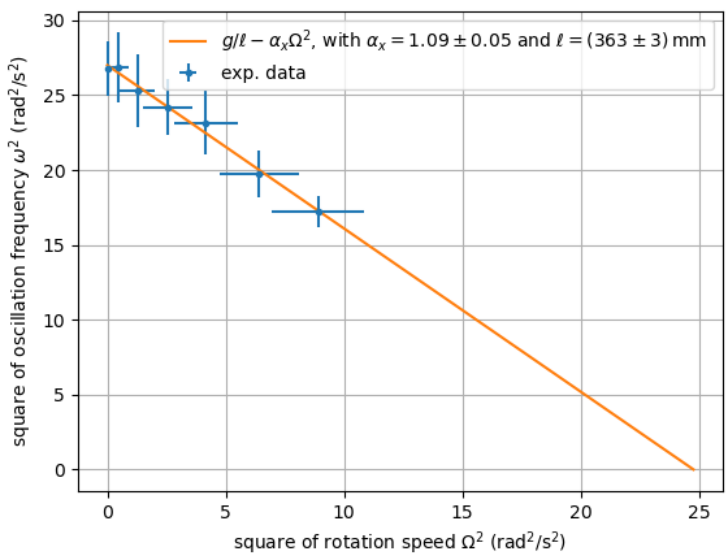
Or, with $\Omega_c^2 = g/l$:

$$\omega^2 = \alpha_x(\Omega_c^2 - \Omega^2)$$

3.10. Using your measurement of the oscillation frequency ω as a function of the rotation speed Ω, and the above equation, graph your data and deduce Ω_c and α_x. Is this dynamic measurement more or less accurate than the static one?

Solution: Graphs of ω vs Ω and their squares for fits of experimental data.





From $\omega^2 = -\alpha_x\Omega^2 + \alpha_x\Omega_c^2$, a straight line in w^2 and Ω^2, and the plot of our experimental data, we find:

$$\alpha_x = 1.09 \pm 0.05$$

From $l = \frac{g}{\alpha_x \Omega_c^2}$, we find:

$$\Omega_c = \sqrt{\frac{g}{\alpha_x l}} = 4.98 \pm 0.1$$

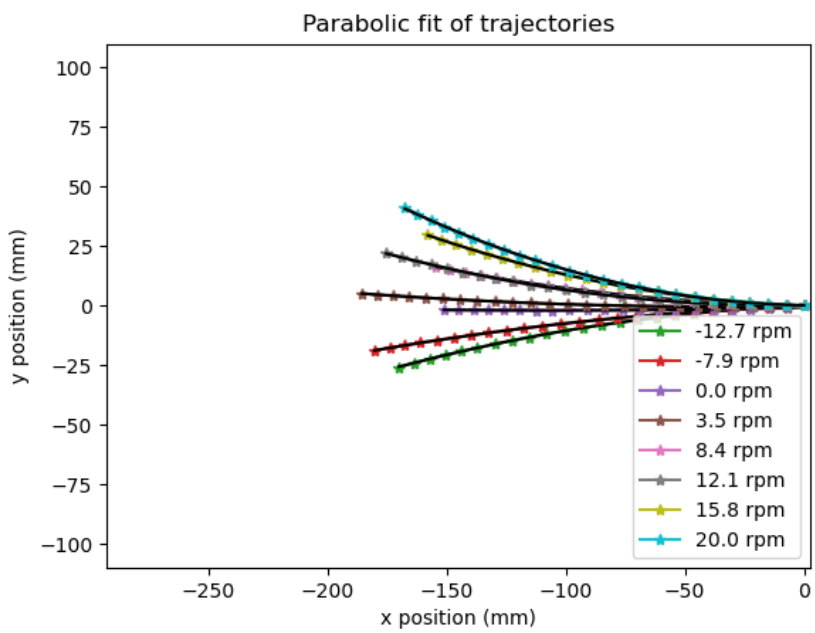
So we have:

$$\Omega_c = 4.98 \pm 0.1; \alpha_x = 1.09 \pm 0.05$$

This result is much more accurate than the one obtained through static analysis of the system, with a standard error of 9% << 38%.

4 Coriolis force: catapult experiment

4.11. Plot all the trajectories on the same graph, and fit them with parabolas. Comment your observations: in which direction are the trajectories turning, depending of the sign of Ω? Deduce the sign of α_v.



Solution: We see on the graph that the curvature of the parabolic fit of the trajectories depend on the sign of the platform's angular velocity Ω. For a positive Ω (defined as counter-clockwise rotation when viewed from above), the trajectories bend toward positive y; for a negative Ω (clockwise rotation), they bend toward negative y. Recall the formula that we found in the dimensional analysis:

$$\vec{F_v} = \alpha_v m \vec{\Omega} \times \vec{v}$$

For $\Omega > 0$, $\vec{F_v}$ must initially point to $+\vec{y}$ and for $\Omega < 0$, $\vec{F_v}$ must initially point to $-\vec{y}$ so by the right-hand rule $a_v < 0$.

4.12. Theoretical analysis: we will assume that the only relevant force to consider in the horizontal plane is the Coriolis one. It is convenient to represent the motion in this plane using a complex representation of the position and speed, $\mathbf{x} = x + iy$, $\mathbf{v} = \dot{x} + i\dot{y}$. Knowing that with this convention $\mathbf{e}_z \wedge \mathbf{v} = i\mathbf{v}$, show that Newton's second law reads:

$$\dot{\mathbf{v}} = i\alpha_v \Omega \mathbf{v}.$$

- The solution of this equation being of the form Ae^{Bt}, with A and B complex coefficients to identify, give the expression of $\mathbf{v}$ and show that the modulus of the speed is constant (we will note this constant v_0).
- Integrate $\mathbf{v}$ and get the expression of $\mathbf{x}$.
- Expand and re-arrange the previous form to get the equation of a circle in the complex plane.
- Express the curvature of the trajectory (inverse of the circle radius) as a function of α_v, Ω and v_0.

Solution: Assuming that the only relevant force to consider in the horizontal plane is the Coriolis one, i.e.

$$\vec{F_v} = \alpha_v m \Omega \vec{z} \times \vec{v}$$

Newton's second law reads as:

$$m\dot{\vec{v}} = \alpha_v m \Omega \vec{z} \times \vec{v}$$

And knowing that $\vec{z} \times \vec{v} = i\vec{v}$, we get

$$\dot{\vec{v}} = i\alpha_v \Omega \vec{v}.$$

This is a first order linear ordinary differential equation, with a solution of the form Ae^{Bt}. We plug this solution in the equation:

$$ABe^{Bt} = i\alpha_v \Omega A e^{Bt}$$

We get $B = i\alpha_v\Omega$, and for $t = 0$, $v(0) = A = v_0$, so $v(t) = v_0 e^{i\alpha_v \Omega t}$.
We find directly that the modulus of $v(t)$ is the constant v_0.
Since $\dot{\vec{x}} = \vec{v}$, we integrate $v(t)$ from 0 to t:

$$x(t) = \int_0^t v(t)dt = \int_0^t v_0 e^{i\alpha_v \Omega t} dt = \frac{v_0}{i\alpha_v \Omega}(e^{i\alpha_v \Omega t} - 1) + x(0)$$

Plugging in again the expression for $v(t)$ we get

$$x(t) = \frac{v(t)}{i\alpha_v \Omega} - \frac{v_0}{i\alpha_v \Omega} + x(0)$$

$$x(t) - (x(0) - \frac{v_0}{i\alpha_v \Omega}) = \frac{v(t)}{i\alpha_v \Omega}$$

$$\left| x(t) - (x(0) - \frac{v_0}{i\alpha_v \Omega}) \right| = \left| \frac{v(t)}{i\alpha_v \Omega} \right|$$

$$\left| x(t) - (x(0) - \frac{v_0}{i\alpha_v \Omega}) \right| = \frac{v_0}{|\alpha_v \Omega|}$$

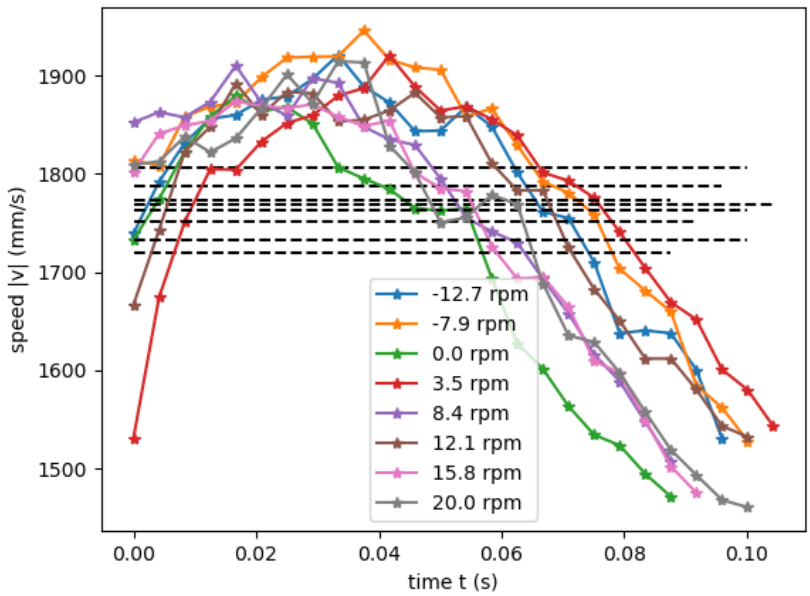
We can write this as:

$$|x(t) - x_c| = R$$

This is the equation of a circle in the complex plane, with $x_c = x(0) - \frac{v_0}{i\alpha_v \Omega}$ the center of the circle and $R = \frac{v_0}{|\alpha_v \Omega|}$ the radius of the circle. Thus the curvature of the trajectory is:

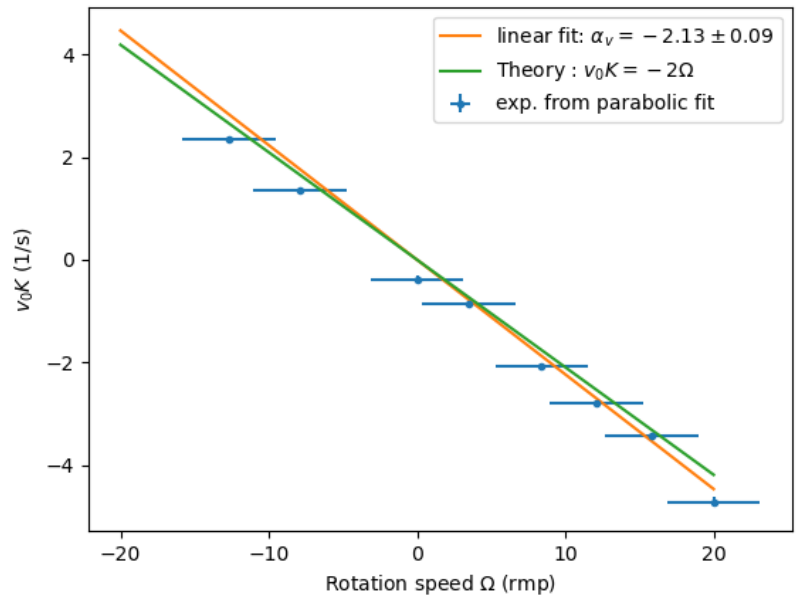
$$\frac{1}{R} = \frac{|\alpha_v \Omega|}{v_0}$$

4.13. Plot the absolute value of the velocity as a function of time of all trajectories in a single graph. Comment the agreement or disagreement with the expectation of a constant modulus.



Solution: In theory, since the Coriolis acceleration is always perpendicular to the velocity, the bead's speed should remain constant over time. The experimental curves show that this is approximately true: all trajectories oscillate between 1700 and 1800 mm/s. However, a slight increase at the beginning and a gradual decrease afterward can be observed, likely due to friction and small energy losses during launch. Also, the launch isn't instantaneous, hence the increase in velocity before 0.02s on the graph. Overall, the results are consistent with the theoretical expectation of a constant velocity modulus, with minor deviations explained by experimental imperfections.

4.14. For a parabola of equation $y = ax^2$, the curvature at the origin is $2a$. Using the parabolic fit of your trajectories, you thus have an estimation of their curvature. Compare this measurement with the theoretical expectation on a graph, and deduce the value of α_v and its uncertainty. How does the value compares with the theoretical one $\alpha_v^{\text{th.}} = -2$?



Solution:

From the linear fit obtained in our plot, we found $a_v^{exp} = -2.13 \pm 0.09$, compared to the theoretical prediction of $a_v^{th} = 2.00$.
Using 95% confidence interval, we find

$$a_v^{exp} \pm 1.96\sigma = -2.13 \pm 0.18.$$

Since $a_v^{th} \in [-2.31; -1.95]$, a_v^{exp} is consistent with the theoretical value.

4.15. Now that we have a quantitative measurement of the Coriolis force, let us justify why we neglected all other contributions. What are the forces acting on the bead during its motion? Why is it reasonable to assume that only the Coriolis force is relevant? You can consider a position $x \sim 10\,\text{cm}$ and a rotation speed $\Omega \sim 20\,\text{rpm} = 2.1\,\text{rad/s}$.

Solution:

Recall the expressions we found for the Coriolis and Centrifugal forces, which are the forces acting on the bead:

$$F_{Cor} \approx m\Omega v, F_{Cent} \approx m\Omega^2 x$$

The average speed value measured was $\approx 1.7 m.s^{-1}$. Taking the ratio of the two forces we get:

$$\frac{F_{Cor}}{F_{Cent}} = \frac{m\Omega v}{m\Omega^2 x} = \frac{v}{\Omega x} = \frac{1.7 m.s^{-1}}{2.1 s^{-1} \times 0.1 m} \approx 8.1$$

The Coriolis force is almost one magnitude bigger than the centrifugal force, which explains why we can neglect it. We can also talk about gravity and air friction. We can neglect gravity because our motion study is confined to the horizontal plane (gravity only acts vertically), and air friction is negligible because the cross sectional area of the bead is very small and therefore the air drag contribution on the total forces acting on the bead is orders of magnitude lower than the Coriolis force.

5 A tornado in the lab

5.16. When falling inwards, the beads acquire spiral-like trajectories. Comment your observations: are the trajectories turning in the same or opposite direction with respect to the sign of Ω? How can you explain the similarity or contradiction with the catapult experiment?

Solution: When the beads fall inward in the tornado setup, their trajectories form spirals that turn in the same direction as the rotation of the platform, counterclockwise for positive Ω and clockwise for negative Ω. This is consistent with what was observed in the catapult experiment. In both setups, the Coriolis force is responsible for the sideways deflection of the bead: since the bead velocity in the rotating frame is directed radially inward, the Coriolis acceleration acts tangentially, bending the trajectory.

In the tornado experiment, the bead is not free but slides along the funnel surface. As it moves inward, gravity pulls it downward, and when it climbs back up the funnel walls, it slows down and reverses direction. This produces a back and forth motion along the slope, resembling a damped harmonic oscillation added on the spiral path.

Thus, both experiments demonstrate the same Coriolis effect: a radially inward motion in a rotating frame is deflected tangentially. The difference in appearance arises from the fact that, in the catapult experiment, the bead is launched freely and deviates once, while in the tornado setup it remains guided by the funnel and oscillates under gravity, creating a continuous spiral pattern.

5.17. Can you draw an analogy between the observed trajectories and the motion of clouds in the atmosphere in a depression (see Fig.2)? Can you predict the direction of the spiral in each hemisphere of the earth?

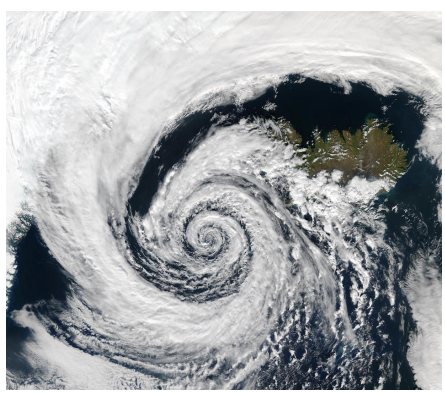


Figure 2: Satellite image of a depression over Iceland.

Solution: First notice that the Earth's rotation is CCW in the Northern hemisphere and CW in the Southern hemisphere. In areas of depression, areas of low pressure in the atmosphere, clouds will naturally form, and analogous to the balls falling down the funnel in a spiral motion, the air from outside the depression will fall radially towards the center of the depression, because of the pressure difference. Due to the Earth's rotation, the air will experience a Coriolis force, which is

predictable since the force acts tangentially to the radial direction and follows the right hand rule, which we can use since we know the Earth's rotation.